Global Pollution Analysis and Energy Recovery Modelling

Machine Learning Project Report

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Course : Machine Learning

1. Introduction

Environmental pollution has become one of the most critical global challenges affecting ecosystems, public health, and economic sustainability. Air pollution, water contamination, soil degradation, and industrial waste generation contribute significantly to environmental deterioration. However, modern waste-to-energy technologies provide an opportunity to convert certain pollutants into usable energy.

The objective of this project is to analyse global pollution data and explore the relationship between pollution indicators and energy recovery potential. Machine learning techniques are used to both predict energy recovery and classify pollution severity levels.

2. Objectives

The main objectives of this study are:

- Analyse pollution indicators across different countries.
- Predict energy recovery using Linear Regression.
- Classify pollution severity using Logistic Regression.
- Evaluate and compare model performance.
- Provide actionable environmental insights.

3. Dataset Description

The dataset contains environmental and economic indicators for multiple countries. The key features include:

Feature	Description
Air_Pollution_Index	**Air pollution level indicator**

Water_Pollution_Index	**Water contamination index**
Soil_Pollution_Index	**Soil pollution measurement**
Industrial_Waste (in tons)	**Industrial waste generated**
Energy_Recovered (in GWh)	**Energy recovered from waste**
CO2_Emissions (in MT)	**Carbon dioxide emissions**
Renewable_Energy (%)	**Percentage of renewable energy usage**
Plastic_Waste_Produced (in tons)	**Plastic waste production**
Energy_Consumption_Per_Capita (in MWh)	**Energy usage per person**
Population(in millions)	**Population size**
GDP_Per_Capita (in USD)	**Economic indicator**
Year	**Observation Year**
Country	**Country Name**

4. Data Preprocessing

Data preprocessing steps included:

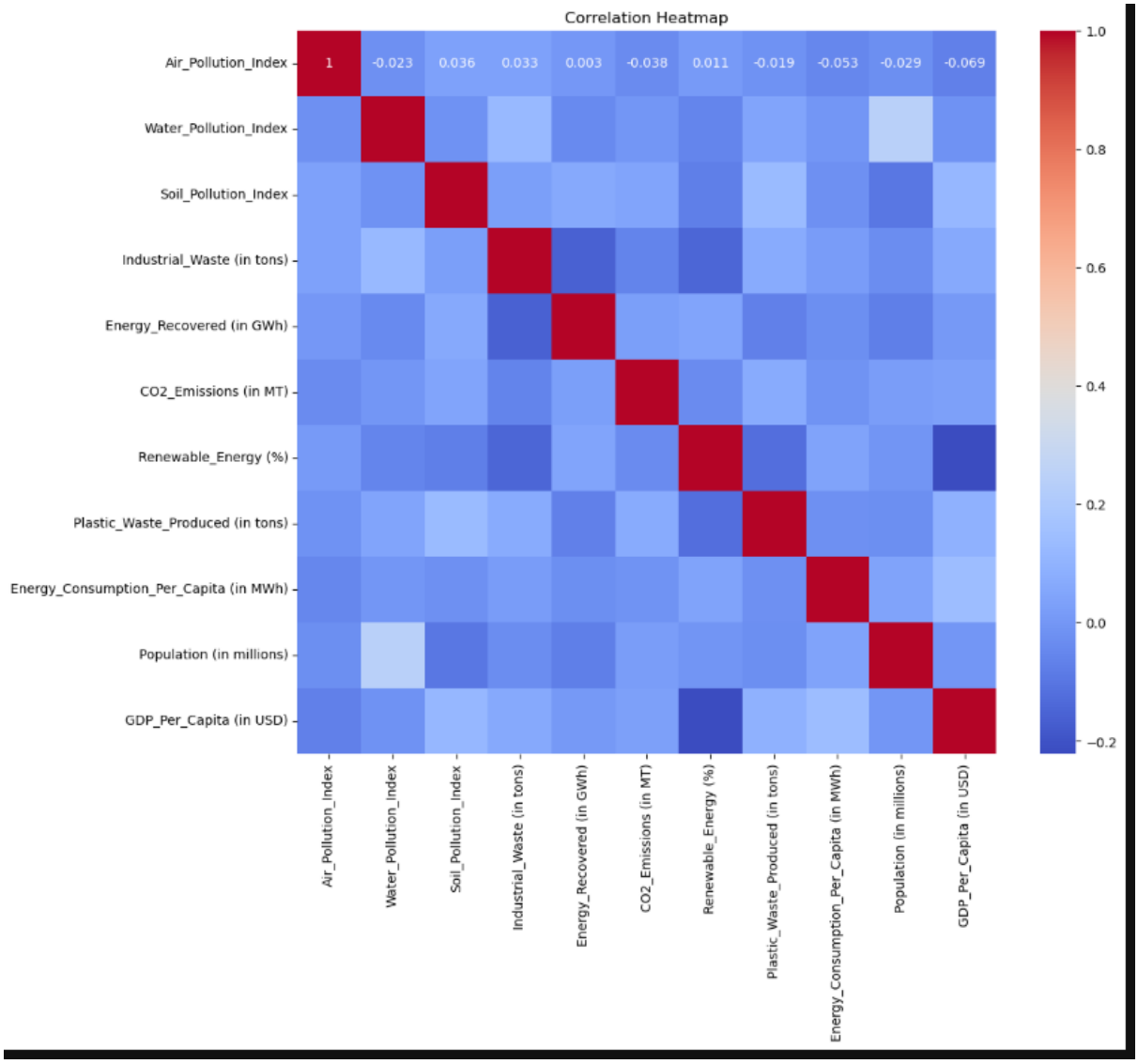
- Handling missing values using appropriate imputation techniques.
- Converting categorical data into numerical format where required.
- Standardizing numerical variables using StandardScaler.
- Removing categorical variables with high cardinality (such as Country) from regression models to avoid dimensional instability.

These steps ensured that the dataset was suitable for machine learning modelling.

5. Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to understand relationships among pollution indicators and energy recovery.

5.1 Correlation Analysis



Correlation Heatmap

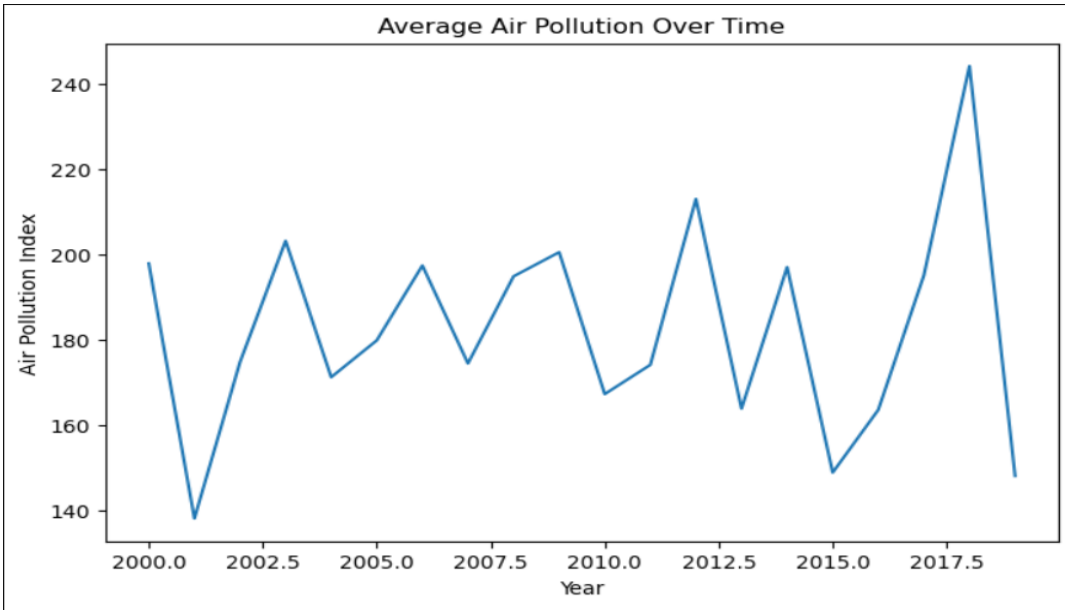
Caption:
Figure 1: Correlation heatmap showing relationships among pollution indicators and energy variables.

Explanation:

The correlation heatmap shows generally weak to moderate relationships between variables. In particular, the correlation between pollution indicators and energy recovery is relatively weak, suggesting that predicting exact energy recovery values using linear relationships may be difficult.

5.2 Pollution Trends Over Time

Line Plot – Air Pollution Trend

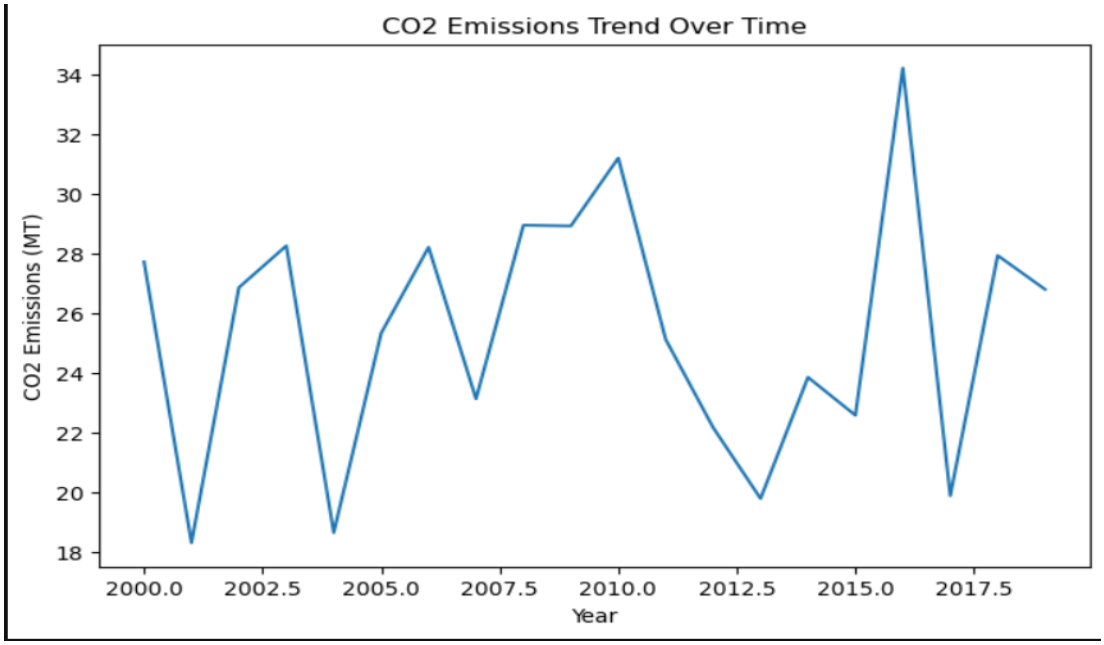


Caption:
Figure 2: Average air pollution index across years.

Explanation:

The line plot illustrates how air pollution levels change over time. Such trends help identify whether pollution levels are increasing or decreasing globally.

Line Plot – CO2 Emissions Trend

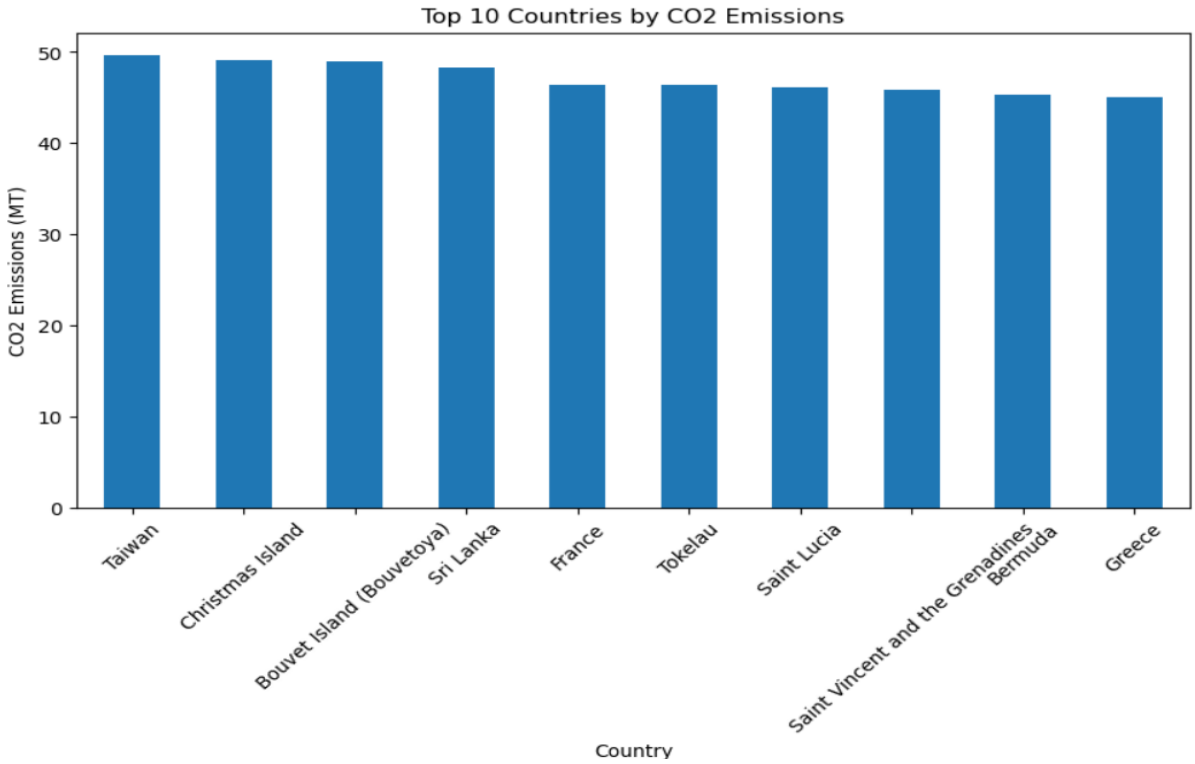


Caption:
Figure 3: CO2 emissions trend over time.

Explanation:

CO2 emissions provide insight into industrial and energy-related environmental impacts.

5.3 Country-Level Pollution Comparison



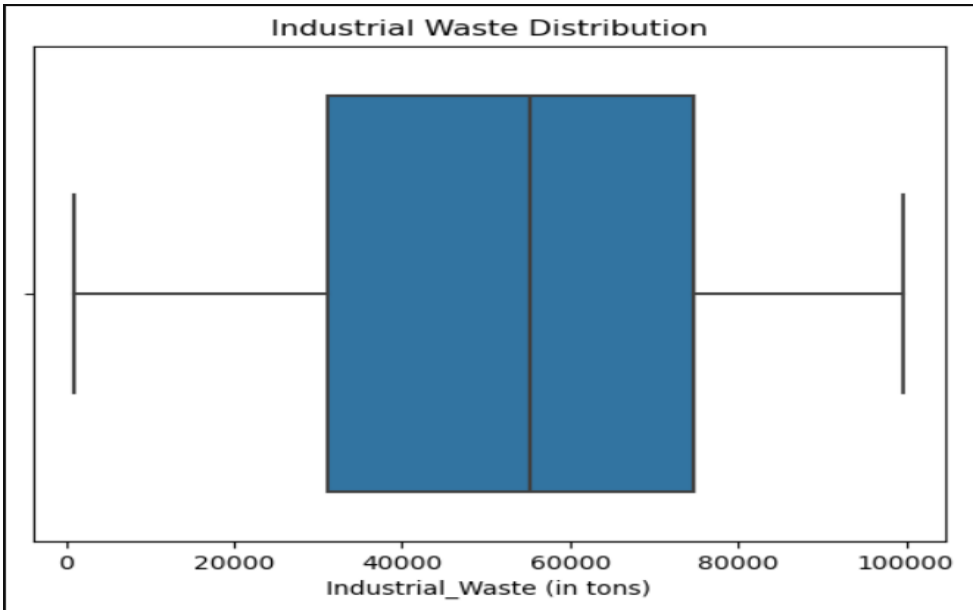
Bar Chart – Top 10 Countries by CO2 Emissions

Caption:
Figure 4: Top 10 countries with highest CO2 emissions.

Explanation:

This visualization highlights countries contributing most to global emissions.

5.4 Distribution of Pollution Indicators



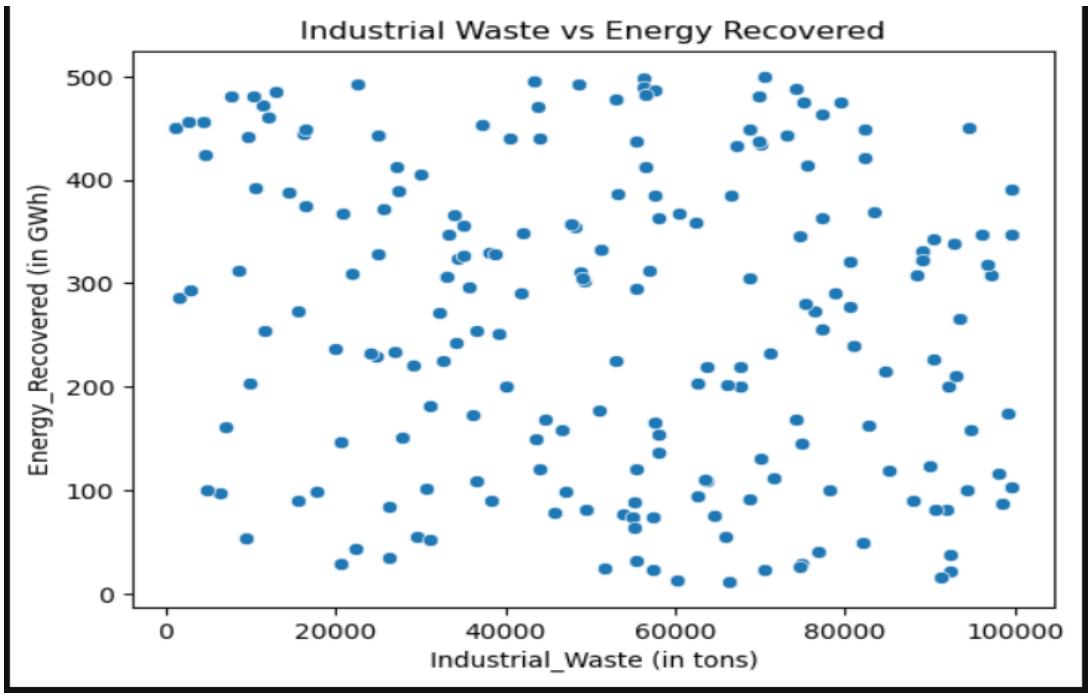
Box Plot – Industrial Waste Distribution

Caption:
Figure 5: Distribution of industrial waste generation.

Explanation:

The box plot identifies potential outliers and the spread of industrial waste values.

Scatter Plot – Industrial Waste vs Energy Recovery



Caption:
Figure 6: Relationship between industrial waste generation and energy recovery.

Explanation:

The scatter plot suggests that although some relationship exists, the relationship is not strongly linear.

6. Feature Engineering

To better analyse pollution severity, a combined pollution indicator was created.

Total Pollution Index was calculated by summing:

- Air Pollution Index
- Water Pollution Index
- Soil Pollution Index

This feature was used to categorize pollution levels into three categories: Low, Medium, and High.

7. Linear Regression Model

7.1 Objective

The objective of the Linear Regression model is to predict **Energy_Recovered (in GWh)** based on environmental indicators such as pollution levels, industrial waste, and emissions.

7.2 Model Performance

Evaluation Metrics:

```
R² Score: -0.4830874468609432
MAE: 0.8292571281791696
```

7.3 Interpretation

The negative R^2 score indicates that the model performs worse than predicting the average energy recovery value. This suggests that the relationship between pollution indicators and energy recovery is weak or potentially nonlinear.

Although pollution contributes to energy recovery potential, other factors such as technology, infrastructure, and energy policies may significantly influence the actual energy recovered.

8. Logistic Regression Model

8.1 Objective

Logistic Regression was implemented to classify pollution severity into three categories:

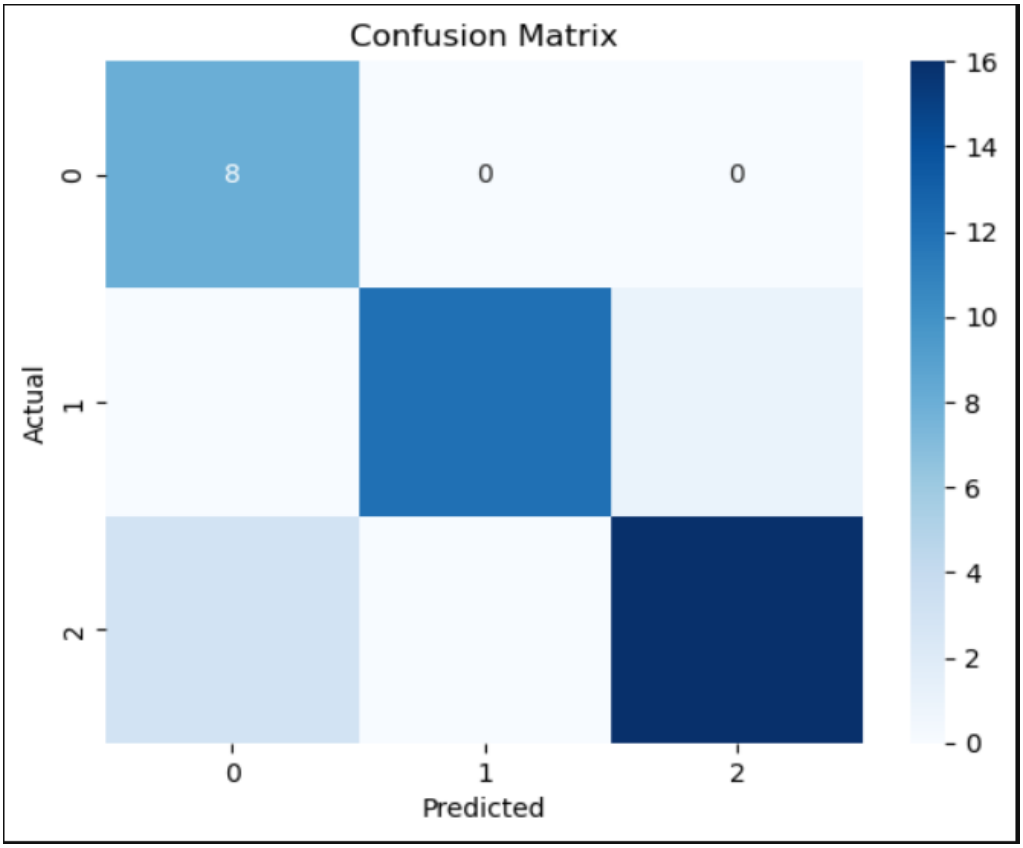
- Low Pollution
- Medium Pollution
- High Pollution

The classification was based on pollution indicators such as air, water, and soil pollution indices.

8.2 Model Performance

Accuracy: **90%**

This indicates that the model successfully classified pollution severity for the majority of countries.




```
print("Accuracy:", accuracy_score(y_test, y_pred))
Accuracy: 0.9
```

Confusion Matrix Caption:
Figure 7: Confusion matrix for pollution severity classification.

Explanation:

The confusion matrix shows that most predictions lie along the diagonal, indicating correct classification of pollution levels.

9. Model Comparison

Model	Task	Performance	Interpretation
Linear Regression	Predict Energy Recovery	$R^2 = -0.15$	Weak predictive ability
Logistic Regression	Classify Pollution Severity	Accuracy = 90%	Strong classification performance

While linear regression struggled to predict exact energy recovery values, logistic regression effectively categorized pollution severity.

10. Actionable Insights

Based on the analysis, several insights were identified:

1. Countries classified under high pollution levels require stronger environmental regulations and monitoring systems.
2. Industrial waste generation plays a significant role in environmental pollution and presents opportunities for waste-to-energy technologies.
3. Increasing renewable energy adoption may help reduce pollution severity and improve sustainability.
4. Governments should invest in advanced waste management systems to improve energy recovery from pollutants.

11. Conclusion

This study analysed global pollution data and applied machine learning techniques to understand the relationship between pollution indicators and energy recovery. Linear Regression demonstrated limited ability to predict energy recovery due to weak linear relationships among variables. However, Logistic Regression successfully classified pollution severity with high accuracy.

These results highlight the importance of effective environmental policies and advanced waste-to-energy technologies to address global pollution challenges.